PlaneWeave: Canonicalizing Bounding Planes for Consistent Video Editing

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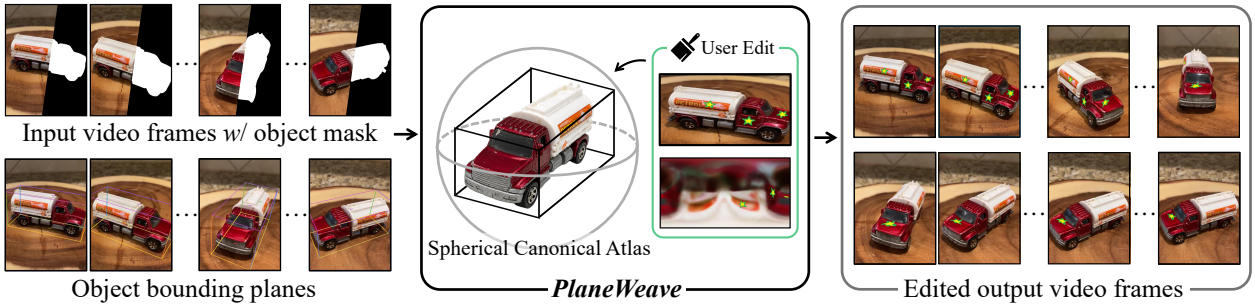


Figure 1. **Overview of *PlaneWeave*.** Given an input video and object bounding planes, *PlaneWeave* establishes a shared spherical canonical atlas. Edits applied once in this atlas are propagated consistently across video frames with large viewpoint changes.

Abstract

Consistent object-level video editing requires a directly editable representation that preserves surface correspondence across frames. Existing atlas-based methods provide intuitive 2D editing domains, but planar representations struggle to maintain consistent surface appearance beyond a limited range of viewpoints. We propose PlaneWeave, a geometry-guided canonicalization framework that maps object observations through a bounding surface to a shared spherical canonical atlas, providing multi-directional surface support within a unified 2D domain. To establish persistent correspondences, PlaneWeave reconstructs sparse 3D anchors from multi-frame tracks and uses their object-centric directions as canonical targets to learn a geometry-guided spatio-temporal weave on a unit sphere S^2. This mapping aligns multi-frame surface observations into a shared spherical atlas, where appearance is jointly reconstructed and edits can be propagated consistently to corresponding object surfaces across frames. Experiments show that PlaneWeave outperforms planar-atlas baselines in terms of cross-frame correspondence accuracy and edit stability under substantial viewpoint changes. For more details, please visit our project page https://vision3d-lab.github.io/planeweave

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1. Introduction

Object-level appearance editing in monocular videos aims to modify the visual appearance of an object, such as color or texture, and propagate the modification consistently throughout the video. This task requires more than independent frame-wise pixel manipulation, as an edit must remain attached to the corresponding physical surface even as different parts of the object surface become visible over time. Despite remarkable progress in neural video editing [6, 20, 30] and dynamic scene representations [13, 14, 29], maintaining such surface-consistent edits remains challenging under large relative viewpoint changes caused by camera motion or object rotation.

Motivated by this challenge, we seek a representation that combines an object-centric 3D scene organization with direct editability. In particular, the desired representation should establish persistent correspondences across frames and provide a persistent 2D appearance domain compatible with standard 2D image editing tools. Existing approaches do not typically offer both within a unified representation. Canonical 2D representations [6, 30] offer intuitive editing interfaces but typically lack explicit object-centric 3D scene organization. In contrast, object-centric 3D representations [10, 19, 28] model object motion and spatial relationships for compositional rendering but do not naturally expose such a 2D appearance domain.

One way to bridge this gap is to combine a directly ed-

itable 2D canonical representation with an object-centric 3D scene organization. Neural Atlas Graphs (NAG) [22] adopts this strategy by associating each object node in a 3D scene graph with a planar neural atlas, which serves as a persistent 2D appearance map shared across video frames. This design allows NAG to handle object motion, depth ordering, and occlusion-aware composition while preserving a 2D editing interface. However, under large relative viewpoint changes, observations from substantially different sides of an object must be organized on the same planar support, making it difficult to maintain distinct surface correspondences within a unified canonical representation.

In this work, we propose *PlaneWeave*, a geometry-guided canonicalization framework that weaves correspondences from different video frames in the shared spherical canonical domain. At its core, *PlaneWeave* constructs a bounding surface around each object using multiple bounded planar faces, referred to as bounding planes, and parameterizes these planes with a shared spherical domain. As illustrated in Fig. 1, these bounding planes serve as a geometric scaffold rather than an exact model of the object surface, providing spatial support for observations from multiple viewing directions. We map the spherical canonical domain to editable 2D atlas domain using either an equirectangular projection (ERP) domain or separate bounding-plane domains, which can then be edited using standard 2D image editing tools.

Building on this geometric scaffold, *PlaneWeave* establishes persistent surface correspondences through geometry-guided weaving. To account for inaccuracies in bounding geometry and object pose estimates, *PlaneWeave* reconstructs 3D anchors from multi-frame correspondence tracks and assigns the object-centric direction of each anchor as a shared canonical target. These geometry-derived targets are used to fit frame-wise spatial weaves for bootstrapping a separate spatio-temporal canonicalization field, which establishes a continuous mapping from video observations to the shared spherical canonical domain. The resulting field is subsequently optimized jointly with a time-independent appearance atlas. Because rendering and editing share the same canonical mapping, an appearance edit specified once can be propagated throughout the video without additional edit-time correspondence estimation. Experiments on CO3Dv2 [21] and Waymo [24] demonstrate improved canonical correspondence consistency and more stable appearance edit propagation than previous baselines, particularly under large relative viewpoint changes.

In summary, our contributions are as follows:

- We propose *PlaneWeave*, a geometry-guided canonicalization framework for consistent video appearance editing that maps bounding-surface observations to a shared spherical canonical atlas, providing multi-directional surface support and direct 2D editability.
- We introduce a two-stage canonical weaving approach guided by shared spherical targets derived from triangulated 3D anchors. Frame-wise spatial weaves bootstrap a shared spatio-temporal weave to establish persistent cross-frame correspondences.
- We evaluate *PlaneWeave* on CO3Dv2 and Waymo under large relative viewpoint changes, demonstrating more accurate cross-frame correspondences and more stable edit propagation over planar-atlas baselines.

2. Related Work

Neural Rendering for Monocular Video. Neural rendering approaches such as NeRF [17] and 3D Gaussian Splatting [7] have enabled photorealistic scene representations and novel-view synthesis. Recent works have extended neural rendering methods to casual monocular captures, addressing challenges arising from weak parallax [2, 4], unreliable cameras [15], and dynamic content [13, 14, 29]. While well suited to reconstruction and rendering, these representations typically do not expose a persistent 2D domain for direct appearance editing. Motivated by this, we propose a representation that combines object-centric 3D scene organization with an editable 2D appearance domain.

Layered and Canonical Video Editing. Classical layered representations [1, 25] decompose videos into depth-ordered components, while neural video decomposition methods [12, 16] further associate layers with objects and their correlated effects. Beyond separating scene components, canonical video representations provide a persistent appearance domain in which edits can be specified once and propagated throughout a video. Layered Neural Atlases (LNA) [6] and Deformable Sprites [30] construct such domains by mapping dynamic content to persistent 2D appearance maps through learned correspondences or deformations. These representations provide intuitive, persistent domains for object-level appearance editing. However, these predominantly 2D canonicalizations lack an explicit object-centric 3D scaffold for organizing observations from multiple object sides under large viewpoint changes. This motivates us to propose a novel representation that combines a persistent, editable 2D domain with object-centric 3D scene organization.

Object-Centric 3D-Grounded Atlases. Object-centric scene representations model a dynamic video as compositions of moving elements, each with distinct 3D support, enabling compositional rendering [9, 19]. Turning per-object 3D supports into editable 2D domains has been explored through multi-chart surface representations [3] and learned surface parameterizations [26, 27]. However, these representations target scene rendering or static shape parameterization rather than a temporally persistent object-level atlas for dynamic video editing. Neural Atlas Graphs (NAG) [22]

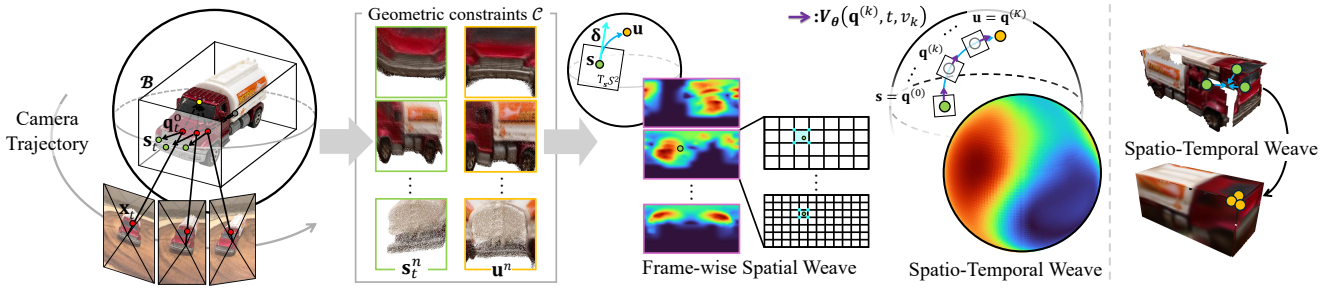


Figure 2. **Overall pipeline of *PlaneWeave***. Given a monocular RGB video with object masks and camera and object poses, correspondences across video observations are aligned in a shared spherical domain. Observations are mapped to initial spherical coordinates from bounding-surface intersections. Spherical canonical targets are estimated and guide the frame-wise and spatio-temporal weaving which further align initialized correspondences.

associate each object node in an object-centric scene graph with a persistent planar neural atlas. However, a single planar atlas can struggle to consistently organize observations from different object sides under large viewpoint changes.

In contrast, *PlaneWeave* maps multiple object-enclosing bounding planes to a shared spherical atlas, represented as an editable equirectangular 2D atlas. This unified atlas representation organizes observations from different object sides for consistent editing throughout the video.

3. Method

PlaneWeave aims to establish a shared canonical appearance domain for consistent object-level appearance editing across frames. Specifically, given a monocular RGB video with object masks and camera and object poses, we learn geometry-guided canonical weaving to align object observations across frames in a shared spherical domain and reconstruct an appearance atlas in this domain.

As illustrated in Fig. 2, we begin with bounding-guided spherical mapping to obtain initial spherical coordinates for object observations (Sec. 3.1). We then perform spherical canonical target estimation using reconstructed 3D anchors (Sec. 3.2). These targets provide supervision for geometry-guided canonical weaving, which aligns observations across frames (Sec. 3.3). Finally, we reconstruct the spherical canonical atlas through joint optimization with the learned weaving field (Sec. 3.4).

3.1. Bounding-Guided Spherical Mapping

We use the object bounding geometry to map object observations to a common object-centric spherical domain. Parameterizing each bounding plane independently would fragment the domain, making it difficult to weave correspondences across plane boundaries. We therefore map bounding-surface intersections to an object-centric unit sphere. To this end, we define the object bounding surface as a union of N bounded planar faces:

$$\mathcal{B} = \bigcup_{i=1}^{N} \mathcal{B}_i, \tag{1}$$

where each $\mathcal{B}_i \subset \mathbb{R}^3$ denotes a bounded planar face of the object bounding volume, expressed in an object coordinate system centered at the bounding volume. The bounding surface serves as a coarse geometric scaffold for organizing object observations rather than an accurate surface model.

For an object pixel $\mathbf{x}$ in frame $t \in \mathcal{T}$, we use the camera and object poses to intersect its back-projected camera ray with the bounding surface:

$$\mathbf{q}_t^o = \Pi_{\mathcal{B}}^t(\mathbf{x}), \tag{2}$$

where $\mathcal{T}$ denotes the set of frame indices, $\Pi_{\mathcal{B}}^t(\cdot)$ maps an image coordinate to the first intersection along its back-projected camera ray with $\mathcal{B}$, and $\mathbf{q}_t^o \in \mathbb{R}^3$ denotes the resulting intersection point in the object coordinate system.

We next map the bounding-surface intersection point onto the unit sphere as

$$\mathbf{s}_t = \rho(\mathbf{q}_t^o) = \frac{\mathbf{q}_t^o}{\|\mathbf{q}_t^o\|_2} \in S^2, \tag{3}$$

where $\rho(\cdot)$ normalizes an object-centric point to unit length, and $\mathbf{s}_t$ denotes the resulting spherical coordinate. Combining the bounding-surface intersection and normalization, we define the mapping from video coordinates to object-centric spherical coordinates as

$$S(\mathbf{x}, t) = \rho\big(\Pi_{\mathcal{B}}^t(\mathbf{x})\big) = \mathbf{s}_t \in S^2. \tag{4}$$

Although S maps observations to a common object-centric spherical domain, it does not guarantee consistent surface correspondences across frames. Because the bounding surface differs from the actual object surface, observations of the same physical surface point from different viewpoints may map to different spherical coordinates, even with accurate object poses. Errors in the bounding geometry and estimated object poses can further increase this discrepancy.

3.2. Spherical Canonical Target Estimation

To address these cross-frame inconsistencies, we estimate shared spherical canonical targets to guide the alignment of object observations through canonical weaving. Cross-frame consistency alone does not determine where observations should map in the canonical domain and admits degenerate solutions, such as mapping all observations to a single point. We therefore derive canonical targets from sparse 3D anchors reconstructed from multi-frame tracks. Specifically, we select well-observed query points and track them across frames. We then triangulate the tracks in the object coordinate system and normalize the resulting 3D points to yield each track's shared spherical canonical target.

Bounding-guided track construction. To initialize tracks in well-observed regions, we select a reference frame r_i for each bounding plane $\mathcal{B}_i$ by maximizing the coverage of projected object observations on that plane. For each bounding plane $\mathcal{B}_i$, we define the coverage score of a binary object mask $\mathbf{M}_t$ in frame t with respect to its fixed grid $\mathcal{G}_i$ as

$$a_i^t = \sum_{\mathbf{g}\in\mathcal{G}_i} \mathbf{1}\left[\mathbf{g} \in \{\Pi_{\mathcal{B}}^t(\mathbf{x}) | \mathbf{M}_t(\mathbf{x}) = 1\}\right], \tag{5}$$

where $\mathbf{1}[\cdot]$ is the indicator function, and the score counts the grid points on $\mathcal{B}_i$ covered by the projected object mask. We then choose $r_i = \arg\max_{t\in\mathcal{T}} a_i^t$. For each reference frame r_i, we sample object pixels whose bounding-surface intersections lie on $\mathcal{B}_i$. We obtain track queries by aggregating these samples across reference frames. We index each track by $n \in \mathcal{N}$, where $\mathcal{N}$ is the set of track indices. We then use the off-the-shelf point tracker [5] to track each query point across the video, obtaining its position $\mathbf{x}_t^n$, visibility, and confidence at each frame t.

Anchor-based canonical target estimation. For each track n, we triangulate a 3D anchor $\mathbf{X}^n$ in the object coordinate system from its tracked positions $\{\mathbf{x}_t^n\}$. During triangulation, we filter outlier tracks to retain valid observations and combine the tracker-provided visibility and confidence scores into a reliability weight w_t^n for each retained observation. We denote the set of valid frame indices for track n by $\mathcal{V}^n \subseteq \mathcal{T}$. Details of triangulation and filtering are provided in the supplementary material.

For each track n, we normalize the reconstructed anchor $\mathbf{X}^n$ to obtain a shared spherical canonical target $\mathbf{u}^n = \rho(\mathbf{X}^n) \in S^2$. We assign this target to the spherical coordinates $\mathbf{s}_t^n = S(\mathbf{x}_t^n, t)$ of all valid observations $t \in \mathcal{V}^n$ in the track (see Fig. 3). These correspondences and their reliability weights define an indexed collection of weighted canonicalization constraints $\mathcal{C}$:

$$\mathcal{C} = \{(\mathbf{s}_t^n, \mathbf{u}^n, w_t^n)\}_{n\in\mathcal{N},\, t\in\mathcal{V}^n}. \tag{6}$$

These constraints guide canonical weaving in Sec. 3.3.

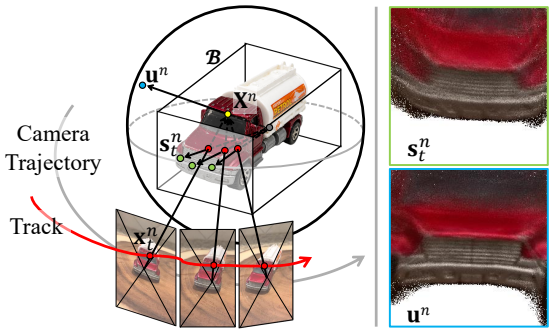


Figure 3. **Illustration of canonical targets**. *Left*: Canonical target derived from a 3D anchor triangulated from a correspondence track. *Right*: Bounding-surface intersections and canonical targets, projected onto one bounding plane for visualization.

3.3. Geometry-Guided Canonical Weaving

Using the weighted canonicalization constraints established in Sec. 3.2, we learn a dense spherical mapping that aligns object observations across frames with their shared canonical targets. We call this mapping a canonical weave. We first construct frame-wise spatial weaves that interpolate sparse geometric constraints into spatially smooth dense mappings. These mappings initialize a shared spatio-temporal weave, which is subsequently refined using canonical target guidance and spatial and temporal regularization.

Frame-wise spatial weave. We first map the bounding-guided spherical coordinates to canonical coordinates independently in each frame. For frame t, we represent this mapping by a spatial weave $F_t : S^2 \to S^2$, constructed from the constraints in $\mathcal{C}$ associated with that frame. To extend these sparse correspondences into a dense mapping while preserving the spherical domain, we express displacements in the tangent space of each input coordinate. The logarithmic map encodes the direction and geodesic distance to the canonical target as a tangent vector, providing displacement samples for interpolation. The exponential map then converts the interpolated displacements back into spherical coordinates. Specifically, each input-target pair $(\mathbf{s}_t^n, \mathbf{u}^n)$ defines a target tangent displacement:

$$\boldsymbol{\delta}_t^n = \mathrm{Log}_{\mathbf{s}_t^n}(\mathbf{u}^n) \in T_{\mathbf{s}_t^n} S^2, \tag{7}$$

where $T_{\mathbf{s}_t^n} S^2$ denotes the tangent space at $\mathbf{s}_t^n$. We interpolate these sparse target displacements to construct a dense tangent displacement field $\boldsymbol{\delta}_t$. Applying the exponential map yields the frame-wise spatial weave:

$$F_t(\mathbf{s}) = \mathrm{Exp}_{\mathbf{s}}\big(\boldsymbol{\delta}_t(\mathbf{s})\big). \tag{8}$$

We construct the dense displacement field using a multi-level B-spline representation [11] over the ERP domain. Starting from a coarse grid, each successive level interpolates the residual displacement left by the preceding lev-

els. At each input spherical coordinate, we sum the interpolated 3D displacement vectors from all levels and project the result onto the tangent plane at that coordinate to obtain $\boldsymbol{\delta}_t(\mathbf{s})$. The resulting frame-wise spatial weaves provide spatially smooth dense mappings guided by sparse geometric constraints, serving as an initialization for a shared spatio-temporal weave that incorporates temporal continuity.

Spatio-temporal weave. We extend the frame-wise spatial alignment into a single time-conditioned mapping, termed a spatio-temporal weave $W_\theta \colon S^2 \times \mathcal{T} \to S^2$. As in NAG [22], we refine geometry-based coordinates through a learned deformation. To reduce the burden of predicting large alignment corrections in a single step, we represent this mapping as a composition of small spherical deformations, with each update evaluated at the current spherical position. We bootstrap W_θ from the frame-wise spatial weaves and subsequently optimize it using canonical target supervision, cycle consistency, and spatial and temporal regularization.

We parameterize the incremental deformation using a coordinate-based MLP $V_\theta : S^2 \times \mathcal{T} \times [0,1] \to \mathbb{R}^3$, implemented with tiny-cuda-nn [18]. At each step k, the MLP predicts a 3D velocity vector from the current spherical position $\mathbf{q}^{(k)}$, video time t, and deformation progress v_k. Starting from $\mathbf{q}^{(0)} = \mathbf{s}$, we update the spherical position using an incremental tangent displacement:

$$\begin{aligned} \mathbf{q}^{(k+1)} &= \mathrm{Exp}_{\mathbf{q}^{(k)}}\big(\boldsymbol{\delta}_t^{(k)}\big), \quad k = 0, \dots, K-1, \\ \boldsymbol{\delta}_t^{(k)} &= \Pi_{\mathbf{q}^{(k)}}\big(\Delta v V_\theta(\mathbf{q}^{(k)}, t, v_k)\big). \end{aligned} \tag{9}$$

Here, K is the number of updates, $\Delta v = 1/K$, and $v_k = k\Delta v$. The operator $\Pi_{\mathbf{q}^{(k)}}(\cdot)$ projects a 3D vector from the MLP scaled by Δv onto the tangent space at $\mathbf{q}^{(k)}$, yielding the small tangent displacement $\boldsymbol{\delta}_t^{(k)}$. After K updates, we obtain the canonical coordinate $W_\theta(\mathbf{s}, t) = \mathbf{q}^{(K)}$.

Bootstrapping from spatial weaves. We first bootstrap the spatio-temporal weave using the spatially smooth mappings provided by the frame-wise weaves. Specifically, we minimize the discrepancy between the spatio-temporal weave and the frame-wise weave outputs at the observations in $\mathcal{C}$:

$$\sum_{n \in \mathcal{N}} \sum_{t \in \mathcal{V}^n} w_t^n \, \psi_{\mathrm{geo}}(g(W_\theta(\mathbf{s}_t^n, t), F_t(\mathbf{s}_t^n))), \tag{10}$$

where $g(\cdot,\cdot)$ denotes spherical geodesic distance and $\psi_{\mathrm{geo}}(\cdot)$ is a robust penalty. This initialization transfers the spatial alignment established by the frame-wise weaves to the shared spatio-temporal mapping.

Canonical target guidance. After bootstrapping, we optimize the spatio-temporal weave using the shared spherical canonical targets estimated in Sec. 3.2. For each observation in $\mathcal{C}$, we penalize the geodesic distance between its mapped coordinate and the corresponding target:

$$\mathcal{L}_{\mathrm{can}} = \sum_{n \in \mathcal{N}} \sum_{t \in \mathcal{V}^n} w_t^n \, \psi_{\mathrm{geo}}(g(W_\theta(\mathbf{s}_t^n, t), \mathbf{u}^n)). \tag{11}$$

Cycle consistency and smoothness. In addition to canonical target guidance, we regularize the weave to encourage forward-reverse consistency and spatial and temporal smoothness. We first impose a cycle-consistency loss that encourages a coordinate to return to its original position after forward and reverse mapping:

$$\mathcal{L}_{\mathrm{cyc}} = \mathbb{E}_{\mathbf{q},t}\Big[g\big(W_\theta^{-1}(W_\theta(\mathbf{q}, t), t), \mathbf{q}\big)^2\Big]. \tag{12}$$

Here, $\mathbf{q} \in S^2$ denotes a sampled spherical coordinate. We approximate the inverse mapping W_θ^{-1} by evaluating V_θ at deformation progress values $v = 1, (K-1)/K, \dots, 1/K$ in reverse order. At each step, we apply a tangent displacement at the current spherical position, scaled by $-\Delta v$, through the exponential map.

To encourage spatially and temporally smooth deformation, we express the overall displacement induced by the weave as $\mathbf{d}_\theta(\mathbf{q}, t) = \mathrm{Log}_{\mathbf{q}}(W_\theta(\mathbf{q}, t))$. We penalize changes in this displacement between neighboring spherical coordinates and neighboring times:

$$\begin{aligned} \mathcal{L}_s &= \mathbb{E}_{\mathbf{q},t,\boldsymbol{\xi}}\left[\frac{\big\|P_{\mathbf{q}' \to \mathbf{q}}\big(\mathbf{d}_\theta(\mathbf{q}', t)\big) - \mathbf{d}_\theta(\mathbf{q}, t)\big\|_2^2}{\epsilon^2}\right], \\ \mathcal{L}_t &= \mathbb{E}_{\mathbf{q},t,t'}\Big[\big\|\mathbf{d}_\theta(\mathbf{q}, t') - \mathbf{d}_\theta(\mathbf{q}, t)\big\|_2^2\Big]. \end{aligned} \tag{13}$$

Here, $\mathbf{q}' = \mathrm{Exp}_{\mathbf{q}}(\epsilon\boldsymbol{\xi})$ is a neighboring spherical coordinate, where $\boldsymbol{\xi} \in T_{\mathbf{q}}S^2$ is a sampled unit tangent direction and ϵ is a small step size. The operator $P_{\mathbf{q}' \to \mathbf{q}}$ parallel-transports the neighboring displacement to $T_{\mathbf{q}}S^2$ so that the two vectors can be compared in the same tangent space. The time t' is sampled near t.

After bootstrapping, we optimize the spatio-temporal weave using the combined objective:

$$\mathcal{L}_{\mathrm{weave}} = \lambda_{\mathrm{can}}\mathcal{L}_{\mathrm{can}} + \lambda_{\mathrm{cyc}}\mathcal{L}_{\mathrm{cyc}} + \lambda_s\mathcal{L}_s + \lambda_t\mathcal{L}_t, \tag{14}$$

where $\lambda_{\mathrm{can}} = 1$ and $\lambda_{\mathrm{cyc}} = \lambda_s = \lambda_t = 10^{-2}$.

3.4. Spherical Canonical Atlas Reconstruction

Using the learned spatio-temporal weave, we reconstruct a spherical canonical appearance atlas from observations across frames. We jointly refine the weave and the atlas through photometric reconstruction while retaining geometric guidance and regularization.

Spherical canonical atlas. We compose the bounding-guided spherical mapping S from Sec. 3.1 with the learned spatio-temporal weave W_θ from Sec. 3.3 to map video coordinates to the spherical canonical domain:

$$Z(\mathbf{x}, t) = W_\theta(S(\mathbf{x}, t), t). \tag{15}$$

We represent canonical appearance using a spherical canonical atlas $A_\phi : S^2 \to [0,1]^3$, shared across frames. For an

object pixel $\mathbf{x}$ in frame t, we reconstruct its RGB color by querying the atlas at its canonical coordinate:

$$\hat{I}_t(\mathbf{x}) = A_\phi(Z(\mathbf{x}, t)), \tag{16}$$

where $\hat{I}_t(\mathbf{x})$ denotes the reconstructed color corresponding to the observed RGB value $I_t(\mathbf{x})$. We supervise appearance reconstruction using a masked photometric loss:

$$\mathcal{L}_{\text{rgb}} = \sum_{t \in \mathcal{T}} \sum_{\mathbf{x}} \mathbf{M}_t(\mathbf{x})\, \psi_{\text{rgb}}\Big(\hat{I}_t(\mathbf{x}) - I_t(\mathbf{x})\Big), \tag{17}$$

where $\psi_{\text{rgb}}(\cdot)$ is a robust photometric penalty.

Joint optimization. Following NAG [22], we jointly optimize the spatio-temporal weave W_θ and the appearance atlas A_ϕ through photometric reconstruction. Although the reconstruction loss provides dense appearance supervision, it does not explicitly enforce consistent canonical coordinates for observations of the same physical surface point. We therefore retain $\mathcal{L}_{\text{weave}}$ and introduce an additional cross-frame correspondence-consistency loss:

$$\mathcal{L}_{\text{corr}} = \sum_{n \in \mathcal{N}} \sum_{\substack{t, t' \in \mathcal{V}^n \\ t < t'}} w_t^n w_{t'}^n\, g(Z(\mathbf{x}_t^n, t), Z(\mathbf{x}_{t'}^n, t'))^2. \tag{18}$$

During joint optimization, we reduce λ_{can} to 10^{-3}, allowing the mapping to refine beyond imperfect anchor-derived targets while preserving cross-frame correspondence consistency. The joint objective is

$$\mathcal{L}_{\text{joint}} = \mathcal{L}_{\text{rgb}} + \mathcal{L}_{\text{corr}} + \mathcal{L}_{\text{weave}}. \tag{19}$$

The reconstructed spherical atlas can be represented in editable 2D domains, including an equirectangular projection (ERP) or separate texture domains for the N bounding planes. Users can inspect and edit the appearance in these domains using standard 2D editing tools. We then propagate edits across frames by querying the edited textures at coordinates determined by the same canonical mapping Z. Details of texture-map construction and edit propagation are provided in the supplementary material.

4. Experiments

4.1. Datasets

To assess cross-view editing consistency, we evaluate *PlaneWeave* on selected sequences of CO3Dv2 [21], where each object-centric capture includes large viewpoint changes. We further evaluate on a preprocessed sequence of Waymo Open Dataset [24] from the NAG [22] test set to assess the performance on a dynamic scene. Details of dataset preprocessing and object bounding-plane initialization are provided in the supplementary material.

4.2. Implementation Details

We implement *PlaneWeave* in PyTorch and train all models on a single NVIDIA RTX 3090 GPU with 24 GB memory. For geometry-guided initialization, we sample up to 3,000 query points per bounding plane and obtain long-range trajectories using AllTracker [5], with the longer image side resized to 768 pixels. The frame-wise spatial weave is represented by a three-level spherical B-spline field [11] with a base resolution of 16×16. Our spatio-temporal weave uses $K = 8$ accumulated tangent updates parameterized by a two-layer MLP with 64 hidden units and degree-4 spherical-harmonic spatial encoding. We train for 1,000 bootstrap, 1,000 spatio-temporal pretraining, 1,000 atlas warm-up, and 10,000 joint optimization steps, sampling 4,096 anchor observations and four video frames per iteration. We use Adam [8] with a learning rate of 10^{-3}, reduced to 10^{-4} for the spatio-temporal weave during joint optimization.

4.3. Evaluation Metrics

Canonical correspondence consistency. LNA, NAG, and *PlaneWeave* map video frame coordinates into canonical points in their atlas domain, implicitly establishing cross-frame correspondences. We evaluate their consistency using held-out AllTracker [5] trajectories.

For a source pixel $\mathbf{x}$ in frame s, we retrieve its target correspondence at frame t among its foreground pixels, whose canonical coordinate is nearest to that of the source point:

$$\hat{\mathbf{y}} = \underset{\mathbf{y}:\ \mathbf{M}_t(\mathbf{y})=1}{\arg\min}\ d_\Phi\big(\Phi(\mathbf{x}), \Phi(\mathbf{y})\big), \tag{20}$$

where Φ is a method-specific canonical mapping from the video coordinates to its canonical space, where $\Phi = Z$ for *PlaneWeave*. Here, d_Φ denotes the method-specific distance in its canonical space. For *PlaneWeave*, we use Euclidean distance between unit vectors in S^2, while we use Euclidean distance in normalized 2D atlas coordinates for the planar atlas baselines. With this predicted correspondence $\hat{\mathbf{y}}$, we measure endpoint error (EPE) between the held-out AllTracker correspondence $\mathbf{y}$.

Local deformation. Even perfect cross-frame correspondence consistency alone does not guarantee the preservation of local structures in the canonical mapping, as observations may still undergo stretching, shearing, or folding, resulting in an atlas that is difficult to edit intuitively.

To measure the local deformation introduced by the spatio-temporal weave W_θ, we estimate a 2×2 Jacobian from neighboring pixel samples. For each sampled object pixel $\mathbf{x}_0$ at frame t, we select four neighboring pixels $\{\mathbf{x}_i\}_{i=1}^4$ offset by 4 pixels along the horizontal and vertical directions. We map these pixels to spherical coordinates $\mathbf{s}_{t,i} = \mathcal{S}(\mathbf{x}_i, t)$ and their canonical coordinates $\mathbf{z}_{t,i} = W_\theta(\mathbf{s}_{t,i}, t)$ for $i = 0, \ldots, 4$. We then estimate the

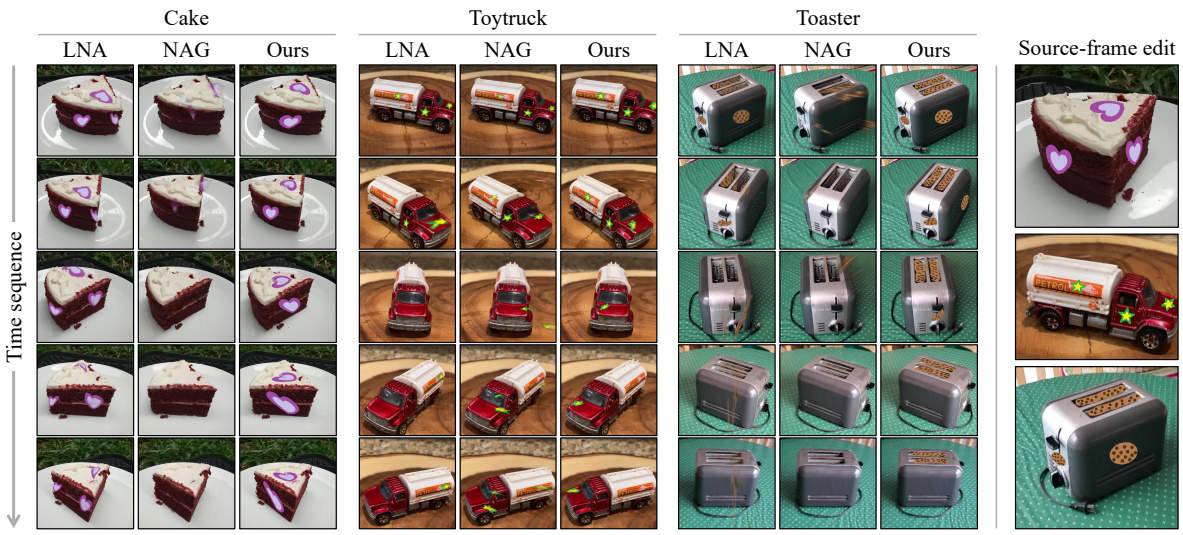


Figure 4. **Qualitative comparisons of consistent appearance editing**. Source-frame edit is propagated across video sequences. *PlaneWeave* shows better editing consistency than LNA and NAG. Object regions are cropped for visualization.

Dataset	Method	Mean EPE ↓	Median EPE ↓	P95 ↓	PCK@5 ↑	PCK@10 ↑
Cake	LNA	86.68	88.95	165.67	0.86	2.81
	NAG	110.73	79.99	294.58	2.29	6.71
	PlaneWeave	**9.90**	**6.34**	**27.12**	**39.21**	**69.89**
Toytruck	LNA	149.11	36.91	584.84	19.79	30.12
	NAG	107.52	61.59	272.47	1.98	7.15
	PlaneWeave	**25.62**	**16.38**	**76.80**	**32.43**	**42.25**
Toaster	LNA	266.86	272.05	566.82	6.19	12.22
	NAG	179.62	135.95	415.76	6.90	9.40
	PlaneWeave	**39.37**	**38.54**	**86.18**	**13.12**	**20.10**

Table 1. **Quantitative correspondence evaluation.** Correspondence accuracy is measured in mean, median, and 95th-percentile (P95) endpoint error (EPE), and percentage of correct keypoints (PCK) at 5- and 10-pixel thresholds. Lower EPE and higher PCK indicate better accuracy. Best results among compared methods are highlighted in **bold**.

local Jacobian $\mathbf{J} \in \mathbb{R}^{2\times 2}$ from the tangent displacements:

$$\mathrm{Log}_{\mathbf{z}_{t,0}}(\mathbf{z}_{t,i}) \approx \mathbf{J}\,\mathrm{Log}_{\mathbf{s}_{t,0}}(\mathbf{s}_{t,i}), \quad i \neq 0, \tag{21}$$

where the displacements are expressed in local orthonormal tangent bases. From $\mathbf{J}$, we derive the following metrics to quantify the local deformation. We define the ARAP [23] error as $\min_{\mathbf{R}\in SO(2)} \|\mathbf{J} - \mathbf{R}\|_F^2$, which measures deviation from local rigidity. From the singular values σ_1, σ_2 of $\mathbf{J}$, we further measure stretch as their deviation from 1, anisotropy as $|\log(\sigma_1/\sigma_2)|$, area distortion as $|\log|\det \mathbf{J}||$, and fold rate as the fraction of samples with $\det \mathbf{J} \leq 0$.

4.4. Comparison

We quantitatively and qualitatively evaluate the correspondence consistency of edits propagated by *PlaneWeave* against planar-atlas baselines LNA [6] and NAG [22] on Cake, Toytruck, and Toaster sequences from CO3Dv2 [21] and a rotating-car sequence from Waymo [24].

Consistent appearance editing. Figure 4 and 6 show source-frame edit propagation across multiple views of the same object. As the viewpoint changes, both LNA and NAG exhibit edit drift and texture distortion. While LNA employs rigidity constraints, NAG uses 3D plane intersections for coarse alignment followed by reprojection-based refinement. Despite these mechanisms, both struggle to maintain consistent canonicalization under large viewpoint changes, suggesting the limitations of relying on rigidity and reprojection-based constraints without explicit canonical targets. In contrast, *PlaneWeave* keeps the edit attached to the corresponding physical surfaces while preserving their spatial extent and shape.

Canonical correspondence consistency. Table 1 further evaluates cross-frame correspondence consistency quantitatively. Overall, *PlaneWeave* outperforms the other baselines on all three sequences by a large margin. In particular, the large reduction in P95 error indicates substantially fewer severe correspondence failures under large viewpoint changes. These results show *PlaneWeave* provides more consistent correspondences than planar-atlas baselines.

4.5. Ablation Study

We ablate three stages of canonicalization by comparing three variants: *Projection*, which uses only the bounding-surface intersection; *Frame-wise*, which additionally applies the frame-wise spatial weave; and *Full*, which further incorporates the spatio-temporal weave. We visualize the ablation on Toaster sequence and compare local deformation across all three sequences. More qualitative comparisons are in the supplementary material.

As shown in Fig. 5 and 6, directly using the bounding-

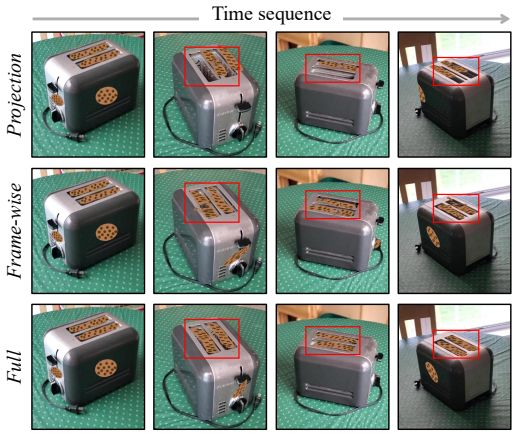


Figure 5. **Effect of plane weaving on Toaster**. Texture propagation across multiple viewpoints is shown for three ablations: *Projection-only*, *Frame-wise*, and *Full*. The red-boxed regions highlight progressively improved temporal consistency and editing quality as additional weaving components are introduced.

Dataset	Model	ARAP ↓	Stretch ↓	Aniso. ↓	Area Dist. ↓	Fold ↓
Cake	*Frame-wise*	0.6910	0.4308	0.3327	0.5101	392 / 188,416
	Full	**0.1325**	**0.2220**	**0.2134**	**0.2991**	**0 / 188,416**
Toytruck	*Frame-wise*	1.1963	0.5529	0.4948	0.5975	679 / 169,984
	Full	**0.3134**	**0.3323**	**0.2744**	**0.4393**	**0 / 169,984**
Toaster	*Frame-wise*	1.9455	0.6584	0.4473	0.7111	1,191 / 169,979
	Full	**0.3426**	**0.3452**	**0.3531**	**0.4099**	**0 / 169,979**

Table 2. **Local deformation integrity.** Local deformation metrics are measured for *Frame-wise* and *Full*. Lower values indicate better preservation of local texture shape.

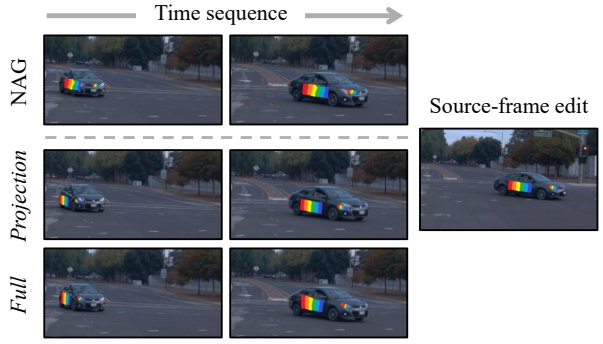


Figure 6. **Consistent editing in a dynamic scene**. At source frame 15, rainbow pattern is placed across the car's right side panel and headlight. This edit is propagated across frames and visualized for selected frames 1, 10 for NAG, *PlaneWeave*'s *Projection* and *Full* variants. LNA is omitted because its inconsistent canonicalization of the dynamic object causes edits to detach from the car.

surface projection leads to noticeable drift of the edited region as the object viewpoint changes. The *Frame-wise* model substantially improves this behavior by aligning observations according to the reconstructed 3D anchors, keeping the edit attached to a more consistent object region across frames. The *Full* model further corrects the remaining local misalignment, producing the most stable edit placement and shape throughout the sequence.



Figure 7. **Editing modalities of *PlaneWeave***. *Top*: Edited videos from canonical atlas editing. *Bottom*: Edited videos from source frame editing.

Table 2 further shows that the spatio-temporal canonicalization improves the local quality of the canonical mapping beyond the frame-wise geometric alignment. Across all three sequences, the *Full* model consistently reduces ARAP, stretch, anisotropy, and area distortion compared with the frame-wise weave, while eliminating all detected folds. Note that the *Projection-only* model is omitted since its near-identity mapping yields trivially near-zero local deformation error. These results indicate that spatio-temporal weave not only improve correspondence consistency but also produces a smoother and less distorted canonical deformation that is more suitable for texture editing.

4.6. Editing Modalities

PlaneWeave supports two complementary interfaces for appearance edits - the observed frame and the spherical canonical atlas. Source-frame editing is intuitive, but it is limited to surfaces visible in the selected frame. Canonical atlas, in contrast, exposes object regions observed across all viewpoints. As shown in Fig. 7, such edits appear consistently when those surfaces become visible later in the sequence.

5. Conclusion

We present *PlaneWeave*, a geometry-guided canonicalization framework for consistent appearance editing under large viewpoint changes. Our method uses bounding surface geometry for coarse alignment and refines it through a spatio-temporal weave guided by sparse tracking correspondences. This establishes more consistent mappings from video observations to a shared spherical canonical atlas than planar-atlas baselines, as demonstrated on static Co3Dv2 and dynamic Waymo sequences. However our method assumes a one-to-one mapping between the object surface and the bounding surface by the intersection with the ray from the object center. This restricts its applicability to approximately convex objects. Future work will extend our framework to objects with more complex geometries.

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